

# Advanced Operational Intelligence through IoT and Big Data analytics

## Business Challenge:

- Client is a leading international mining company with operations worldwide.
- Client faced problems with unscheduled downtime of assets leading to a loss of productivity
- Increased wear and tear on the equipment due to lack of timely preventive maintenance resulting in engine failures
- Lack of alarm stratification :On an average, 173K alarm instances were generated by the Micro-Electro-Mechanical (MEM) system. However, there was a lack of instance prioritization.

## Solution:

- Application of an extensive array of 200 sensors per truck across 300 trucks at 12 mining sites, generating 60,000 readings/second.
- Leveraged and extended existing EDW infrastructure, setting up a parallel Hadoop environment.
- Hexaware used R, for the statistical analysis of the data and creating varied machine learning models.
- Created analytical models for predictive maintenance
- Random Forest Clustering was done to identify high, medium, and low risk scenarios for engine failures and their indicative features.

**\$0.9M**

Potential savings by decrease in critical alarms

**\$1.2M**

Estimated savings by preventing excessive idling

**\$5.5M**

Estimated savings by predictive maintenance of haul truck engines

**\$5.2M**

Potential savings from prolonging engine life

**\$13Mn**

Estimated Savings

